

HCC Coding Gap Classifier

A fine-tuned small language model that flags Medicare risk-adjustment documentation gaps — built to prove a small, locally-hosted model can match or beat a frontier LLM on a narrow enterprise task, at a fraction of the cost.

The Problem

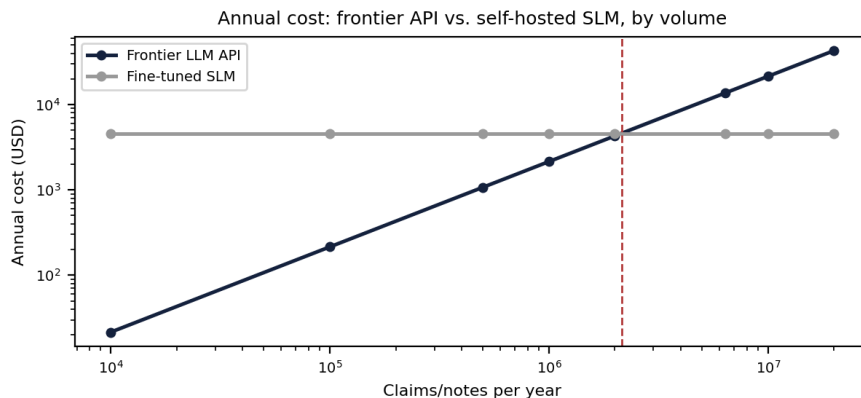
Medicare pays insurers more for sicker patients through the CMS-HCC risk adjustment model: each diagnosis maps to a Hierarchical Condition Category (HCC) with a real dollar value. If a clinician's note doesn't document enough clinical specificity to justify a billed HCC, that's a documentation gap — lost revenue or audit exposure. Insurers run large human chart-review teams to catch this at scale. This project automates the first pass of that review.

Headline Result — Held-Out Test (162 examples, never used to guide any tuning decision)

Metric	Claude Opus 4.7 (frontier, prompted)	Fine-tuned Phi-3.5-mini (SLM, LoRA)
Accuracy	75.9%	88.9%
Cost per note	\$0.0021	\$0.00 marginal
Avg latency	1.15 sec	1.10 sec
Latency std dev	0.77 sec	0.07 sec

The fine-tuned SLM outperformed the prompted frontier model on accuracy, cost, and latency consistency.

ROI: Cost at Scale



Using a real AWS g4dn.xlarge hosting rate (\$0.526/hr) against the measured Claude API cost per note, the breakeven point is **~2.15 million notes/year**. UnitedHealthcare's ACA marketplace segment alone processed 6.4 million claims in 2024 (CMS Transparency in Coverage data) — 3x past breakeven, on just one product line out of many across a 49M+ member book of business.

Methodology

- Ground truth: official CMS-HCC Model V28 ICD-10-to-HCC crosswalk and RAF dollar-value coefficients, sourced from CMS's own published model data.
- Realistic population structure: CMS DE-SynPUF synthetic Medicare claims (10,000 beneficiaries), real chronic-condition flags and demographics.
- Synthetic clinical notes generated at varying documentation specificity (Fully Supported / Insufficient / Unsupported), anchored to real diagnosis codes — real charts are never public.
- Hard-negative training examples added after error analysis revealed a relevance-checking blind spot (wrong disease subtype, normal/resolved values, family history vs. patient history).
- Final 162-example held-out test set built independently and inspected exactly once, after the model was already locked in — preventing test-set leakage from iterative development.
- LoRA fine-tuning of Phi-3.5-mini-instruct (4-bit) via Apple's MLX framework, entirely local, no cloud GPU required for training.

Known Limitations

Clinical note text is synthetic, not drawn from real charts (labels are grounded in the real CMS-HCC crosswalk, not the note text itself). The model still struggles on adversarial cases requiring cross-referencing a specific clinical value against the billed code — these are the cases that should escalate to human review in production, not be auto-resolved.